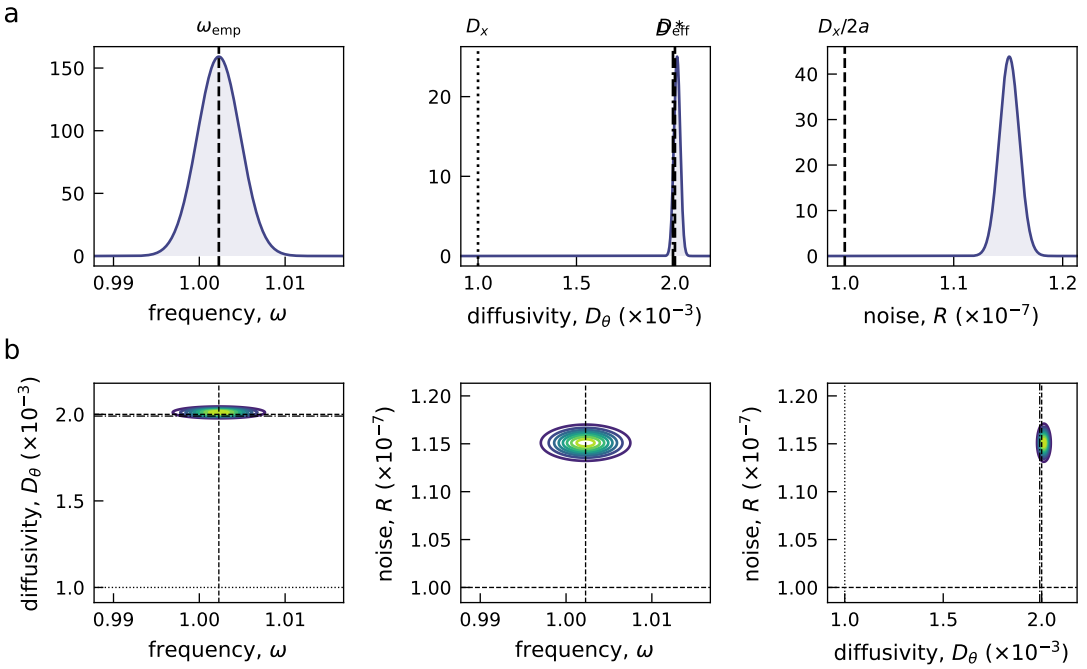

a
ω_{emp}
150
100
50
0
0.99
1.00
1.01
frequency, ω
D_x
D^*_{eff}
20
10
0
1.0
1.5
2.0
diffusivity, D_θ ($\times 10^{-3}$)
$D_x/2a$
40
30
20
10
0
1.0
1.1
1.2
noise, R ($\times 10^{-7}$)
b
diffusivity, D_θ ($\times 10^{-3}$)
2.0
1.5
1.0
0.99
1.00
1.01
frequency, ω
noise, R ($\times 10^{-7}$)
1.20
1.15
1.10
1.05
1.00
0.99
1.00
1.01
frequency, ω
noise, R ($\times 10^{-7}$)
1.20
1.15
1.10
1.05
1.00
1.0
1.5
2.0
diffusivity, D_θ ($\times 10^{-3}$)